OBJECTIVES

- To introduce & define the topic
- To explain types of encephalitis
- To enlist risk factors & etiology
- To explain clinical manifestation & diagnostic evaluation
- To explain management & prevention of encephalitis.

INTRODUCTION

- Encephalitis is a rare yet serious disease that can be life-threatening.
- Encephalitis is an inflammation of the brain tissue.
- The most common cause is viral infections.
- In rare cases it can be caused by bacteria or even fungi.

DEFINITION

- Encephalitis is an inflammation of the brain tissue.

TYPES

- Primary encephalitis- It occurs when a virus directly infects the brain and spinal cord.
- Secondary encephalitis- It occurs when an infection starts elsewhere in the body and then travels to your brain.

RISK FACTORS

- Older adults
- Children under the age of 1 year
- People with weak immune systems

ETIOLOGY

- **Primary (infectious) encephalitis**
- Common viruses, including HSV (herpes simplex virus) and EBV (Epstein-Barr virus)
- Childhood viruses, including measles and mumps
- Arboviruses (spread by mosquitoes, ticks, and other insects), including Japanese encephalitis, West Nile encephalitis, and tick-borne encephalitis

- **Secondary encephalitis:** could be caused by a complication of a viral infection.

CLINICAL MANIFESTATIONS

- Fever
- Headache
- Vomiting
- Stiff neck
- Lethargy (exhaustion)
- Confusion
- Drowsiness
- Hallucinations
- Slower movements
- Coma
- Seizures
- Irritability
- Sensitivity to light
- Unconsciousness

PATHOPHYSIOLOGY

Diffuse cerebral edema, congestion and hemmorrhages

Necrosis and degeneration of neurons

Meningeal congestion with mononuclear infiltration, perivascular tissue necrosis and myelin breakdown

Glial proliferation

type of infecting agent

Demyelination, vascular and perivascular destruction and cerebral cortical involvement

In case of rabies and herpes simplex infection, specific inclusions are identified. Characteristic pathological changes are found in Falciparum malaria.

DIAGNOSTIC EVALUATION

- Careful health history and physical examination
- CSF study helps to differentiate the condition from meningitis
- Blood examination for sugar, urea, electrolytes and metabolic products
- Urine examination, toxicologic study and virological study
- CT Scan
- EEG

COMPLICATIONS

- Loss of memory
- Behavioral/personality changes
- Epilepsy
- Fatigue
- Physical weakness
- Intellectual disability
- Lack of muscle coordination
- Vision problems
- Hearing problems
- Speaking issues
- Coma
- Difficulty breathing
- Death

MANAGEMENT

- **A. MEDICAL MANAGEMENT**
- **I. PHARMACOLOGICAL MANAGEMENT**
- **Anti-inflammatory drugs & antipyretics** - acetaminophen (Tylenol, others), ibuprofen (Advil, Motrin IB, others) and naproxen sodium (Aleve) — to relieve headaches and fever
- **Antiviral drugs**
- Acyclovir (Zovirax)
- Ganciclovir (Cytovene)

- Mannitol or glycerol may be needed to reduce ICP.
- Corticosteroid (dexamethasone)
- Antibiotics
- Vitamin and mineral supplementation
- IV fluid therapy and dopamine to be given to treat shock and fluid-electrolyte imbalance.
- Anticonvulsive drugs

- **II. NON- PHARMACOLOGICAL MANAGEMENT**
- Oxygenation to be provided by nasal cannula
- Mechanical ventilation is necessary in cardio-respiratory insufficiency.

- **NURSING MANAGEMENT**
- **ASSESSMENT**
- Obtain a history of recent infections such as upper respiratory infection, and exposure to causative agents.
- Assess neurologic status and vital signs.
- Evaluate for signs of meningeal irritation.
- Assess sensorineural hearing loss (vision and hearing), cranial nerve damage (eg, facial nerve palsy), and diminished cognitive function.

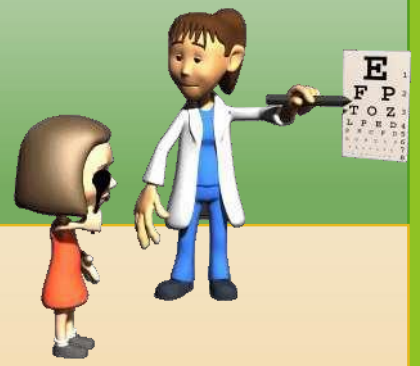
DIAGNOSIS

- Ineffective Tissue Perfusion (cerebral) related to infectious process and cerebral edema
- Risk for Imbalanced Fluid Volume related to fever and decreased intake
- Hyperthermia related to the infectious process and cerebral edema
- Acute Pain related to meningeal irritation
- Impaired Physical Mobility related to prolonged bed rest

GOAL

- To Enhanced Cerebral Tissue Perfusion
- To Maintain Fluid Balance
- To Reduce Fever
- To Reduce Pain
- To Return to Optimal Level of Functioning/ mobility

INTERVENTIONS



- **Enhancing Cerebral Perfusion**
- Assess LOC, vital signs, and neurologic parameters frequently. Observe for signs and symptoms of ICP (eg, decreased LOC, dilated pupils, widening pulse pressure).
- Maintain a quiet, calm environment to prevent agitation, which may cause an increased ICP.
- Prepare patient for a lumbar puncture for CSF evaluation, and repeat spinal tap, if indicated. Lumbar puncture typically precedes neuroimaging
- Notify the health care provider of signs of deterioration: increasing temperature, decreasing LOC, seizure activity, or altered respirations.

- **Maintaining Fluid Balance**
- Prevent I.V. fluid overload, which may worsen cerebral edema.
- Monitor intake and output closely.
- Monitor CVP frequently.
- Osmotic diuretic administration

- **Reducing Fever**
- Administer antimicrobial agents on time to maintain optimal blood levels.
- Monitor temperature frequently or continuously, and administer antipyretics as ordered.
- Institute other cooling measures, such as a hypothermia blanket, as indicated.

- **Reducing Pain**
- Administer analgesics as ordered; monitor for response and adverse reactions. Avoid opioids, which may mask a decreasing LOC.
- Darken the room if photophobia is present.
- Assist with position of comfort for neck stiffness, and turn patient slowly and carefully with head and neck in alignment.
- Elevate the head of the bed to decrease ICP and reduce pain.

- **Promoting Return to Optimal Level of Functioning**
- Implement rehabilitation interventions after admission (eg, turning, positioning).
- Progress from passive to active exercises based on the patient's neurologic status.

- **EXPECTED OUTCOMES**
- Enhanced Cerebral Tissue Perfusion
- Reduced Fever
- Fluid Balance Maintained
- Reduced Pain
- Return to Optimal Level of Functioning

PREVENTION

- Practice good hygiene.
- Don't share utensils.
- Get vaccinations.
- Protection against mosquitoes and ticks
- Dress to protect yourself.
- Apply mosquito repellent.
- Use insecticide.
- Avoid mosquitoes.
- Get rid of water sources outside your home.
- Look for outdoor signs of viral disease- sick or dying birds or animals.

SUMMARY

- We have discussed about encephalitis, types, its cause, risk factor, pathophysiology, clinical manifestation, diagnostic evaluation & management & prevention.

CONCLUSION

- Encephalitis is an inflammation of the brain. Usually the cause is a viral infection, but bacteria can also cause it. It can be mild or severe. Most cases are mild. People may have flu-like symptoms.

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THANK
YOU!